

3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0

∴ The condition that for each column having a leading entry, all positions below the leading entry position must contain 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 2 = 0

∴ The condition that for each column having a leading entry, all positions above the leading entry position must contain 0 is True.

∴ The matrix $\begin{bmatrix} 1 & 0 & 3/2 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix}$ is in rref. $\square$

All the columns of A are pivot columns.

∴ By defn, there are no free variables. ∴ By Theorem 2, this system of equations has a unique solution

d) The range of the transformation T is the set of all linear combinations of the columns of A.

We need to find out whether for some $c_1, c_2 \in \mathbb{R}$, $c_1 \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$

By Theorem 3, this vector equation has the same solution as the system of equations corresponding to the augmented matrix $\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix}$

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \xrightarrow[\text{-3} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ -1 & 7 & 5 \end{bmatrix} \xrightarrow[\text{+ Row 1}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{bmatrix} \xrightarrow[\times 1/14]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 4 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 4 & 8 \end{bmatrix} \xrightarrow[\times 1/4]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow[\text{- Row 2}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 5/2 \end{bmatrix}$$

Verification of whether the final matrix is in echelon form

1. All three rows of the matrix are non-zero rows. ∴ The condition that all the non-zero rows must be above the zero rows is True.

2. Row 1 Leading entry = 1 in Column 1

Row 2 Leading entry = 1 in Column 2

Row 3 Leading entry = 5/2 in Column 3

∴ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.